

Technical Requirement Document (TRD)

System Architecture

PDF ↓ Text Extraction ↓ Chunking ↓ Embedding Generation ↓ Qdrant Vector Database ↓ Similarity Search ↓ Gemini 2.5 Flash ↓ Answer + Citations

Technology Stack

Backend

FastAPI

Vector Database

Qdrant

Embedding Model

BAAI/bge-small-en-v1.5

LLM

Gemini 2.5 Flash

PDF Processing

PyPDFLoader

Chunking

RecursiveCharacterTextSplitter

Embedding Framework

SentenceTransformers

API Testing

Swagger UI

Chunking Configuration

Chunk Size:

1000 characters

Chunk Overlap:

200 characters

Embedding Dimension

384

Vector Search Parameters

Similarity Metric:

Cosine Similarity

Top-k Retrieval:

5

API Endpoints

POST /upload

Uploads a PDF.

Input:

PDF file

Output:

Upload success status

POST /query

Input:

Question

Output:

Answer and source chunks

Core Modules

Document Loader Module

Responsibility:

Extract text from PDFs.

Chunking Module

Responsibility:

Create overlapping chunks.

Embedding Module

Responsibility:

Convert chunks into vectors.

Vector Store Module

Responsibility:

Persist embeddings inside Qdrant.

Retrieval Module

Responsibility:

Retrieve top-k relevant chunks.

Generation Module

Responsibility:

Generate answers using Gemini.

Citation Module

Responsibility:

Return supporting chunks.

Folder Structure

project/

backend/

app/

loaders/

chunkers/

embeddings/

vectordb/

retrievers/

llm/

api/

models/

utils/

requirements.txt

main.py

Future Extensibility

Phase 2:

Hybrid Search

Phase 3:

Reranking

Phase 4:

ReAct Agent

Phase 5:

Self Verification

Phase 6:

Memory

Phase 7:

Workspace System